TEST REPORT

EN IEC 61496-1:2020

Safety of machinery - Electro-sensitive protective equipment -

Part 1: General requirements and tests

Report Reference No. ..... :	SHES230601129701-02
Date of issue ..... :	2023-09-28
Total number of pages ..... :	65
Tested by (name + signature) ..... :	Nino Sun Nino Sun
Approved by (name + signature) ..... :	Lucy Wang Lucy Wf
Testing Laboratory ..... :	SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.
Address ..... :	588 West Jindu Road, Xinqiao, Songjiang, 201612 Shanghai, China.
Applicant's name ..... :	Richbeam (Suzhou) Technology Co., Ltd.
Address ..... :	No.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Test specification	
Standard ..... :	EN IEC 61496-1:2020
Test procedure ..... :	SGS-CSTC
Master TRF ..... :	N/A
Test item description ..... :	2D TOF LIDAR
Trade Mark ..... :	Richbeam
Manufacture ..... :	Same as applicant
Model/Type reference ..... :	LoraBeam1, LoraBeam1H, LoraBeam1S, KoraBeam1, KoraBeam1S
Ratings ..... :	See nameplate

Summary of testing: All tests and assessments are performed on the prototype of the original sampled machine (LoraBeam1) and the technical file which are submitted by the client. Considering the results of the performed test according to EN IEC 61496-1:2020, the items under test are IN COMPLIANCE with the requested specifications specified in the standard.

Tests performed (name of test and test clause):	**Test location:**
5.2.1.2.2 Influence of incandescent light 5.2.1.2.3 Influence of incandescent light reflected by the background 5.2.1.3 Endurance test of the detection capability 5.4.2.2 Ambient temperature variation and humidity test procedure 5.4.2.3 Condensing test procedure 5.4.4.3 Change of temperature 5.4.6.4 Light interference-Incandescent light 5.4.6.5.2 Light interference-Fluorescent light 5.4.6.5.3 Light interference-Fluorescent light (Fail to danger) 5.4.6.6 Light interference-Flashing beacon 5.4.6.7 Light interference-Laser beam pointer 5.4.7.2 Pollution test with opaque test spot 5.4.7.3 Test of homogeneous pollution of the emitter and receiver aperture 5.4.8.1 Background interference 5.4.9.1 Test with opaque test spots 5.4.9.2 Manual interference test with AOPDDR covered 5.4.9.3 Manual interference test in the zone with limited detection capability 5.4.10 Optical occlusion in the detection zone (eclipsed by small object)	Tested in Jining Institute of Quality & Metrology Inspection
5.4.4.1 Vibration 5.4.4.2 Shock 5.4.4.4 Hammer tests 5.4.5 Enclosures	Tested in SGS-CSTC Standards Technical Services (Shanghai) Co., Ltd.
5.4.3.1 Supply voltage variations 5.4.3.2 Supply voltage dips and interruptions 5.4.3.3 Burst 5.4.3.4 Surge 5.4.3.5 Electromagnetic field 5.4.3.6 Conducted disturbances induced by radio-frequency fields 5.4.3.7 Electrostatic discharge	Tested in Compliance Certification Services (Kunshan) Inc.

Copy of marking plate
(The artwork below may be only a draft.)

PR：2D TOF LIDAR
PML：KoraBeam1
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time：Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：KoraBeam1S
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time：Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：LoraBeam1
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time：Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：LoraBeam1H
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time：Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

PR：2D TOF LIDAR
PML：LoraBeam1S
S/N：RB001A2XXYYZZ
MAC：A1:B2:C3:XX:YY:ZZ
MFG：10/02/2023
Richbeam(Suzhou)Technology Co.,LTD
Power Input：(9 ... 36) V DC（Typ. 2.5W）
Aperture Angle：270°（Horizontal）
Scanning Range：Max(2...10)m
(AT 2% Remission)
Response Time：Typ. 33 ms
Class 1 Laser Product
IP67
Type 1
MADE IN CHINA

Test item particulars ..... :	
Classification of installation and use ..... :	Class I
Supply Connection ..... :	Fixed
Possible test case verdicts:	
- test case does not apply to the test object..... :	N/A
- test object does meet the requirement..... :	Pass
- test object does not meet the requirement ..... :	Fail
Testing ..... :	
Date of receipt of test item ..... :	2023-06-21
Date (s) of performance of tests ..... :	2023-06-21 to 2023-08-24

General remarks:

The test results presented in this report relate only to the object tested.

This report shall not be reproduced, except in full, without the written approval of the testing laboratory.

"(see Enclosure #)" refers to additional information appended to the report.

"(see appended table)" refers to a table appended to the report.

Throughout this report, a point is used as the decimal separator.

This document is issued by the Company subject to its General Conditions of Service printed overleaf, available on request or accessible at http://www.sgs.com/en/Terms-and-Conditions.aspx and, for electronic format documents, subject to Terms and Conditions for Electronic Documents at http://www.sgs.com/en/Terms-and-Conditions/Terms-e-Document.aspx. Attention is drawn to the limitation of liability, indemnification and jurisdiction issues defined therein.

Any holder of this document is advised that information contained hereon reflects the Company's findings at the time of its intervention only and within the limits of Client's instructions, if any. The Company's sole responsibility is to its Client and this document does not exonerate parties to a transaction from exercising all their rights and obligations under the transaction documents. This document cannot be reproduced except in full, without prior written approval of the Company. Any unauthorized alteration, forgery or falsification of the content or appearance of this document is unlawful and offenders may be prosecuted to the fullest extent of the law.

Unless otherwise stated the results shown in this test report refer only to the sample(s) tested and such sample(s) are retained for 30 days only.

Manufacturer's Declaration:	
The application for obtaining a Test Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided..... :	☐ **Yes** ☒ **Not applicable**

When differences exist; they shall be identified in the General product information section.	
Name and address of factory (ies) .....:	N/A
General product information and other remarks: This product is 2D TOF LIDAR. Ranging type radar, no OSSDs. In the test, monitoring the output distance instead of the status of the OSSD. Except that the below deviation, are identical with the original product as follows: Performance of the measured distance which is limited by the firmware. All the type variants (models) were assessed and found in conformity with the Machinery Directive requirements	

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
4	**Functional, design and environmental requirements**		—
4.1	Functional requirements		—
4.1.1	Normal operation		-
	Normal operation is the state of an ESPE where no faults are detected and where the OSSD(s) are allowed to be in the ON-state or the OFF-state depending on the state of the sensing function and operating mode.	In the test, monitoring the output distance instead of the status of the OSSD	P
	In normal operation, the ESPE shall respond by giving (an) appropriate output signal(s) when part of a person greater than or equal to the detection capability (as specified in the relevant part of IEC 61496) enters or is in the detection zone.		P
	The ESPE response time shall not exceed that stated by the supplier. No means of adjustment of the response time shall be possible without the use of a key, keyword or tool.		P
4.1.2	Sensing function		-
	The detection capability shall be effective over the detection zone specified by the supplier. No adjustment of the detection zone, detection capability or blanking function (monitored, unmonitored, fixed or floating) shall be possible without the use of a key, keyword or tool.		P
4.1.3	Types of ESPE		-
	In this document, three types of ESPEs are considered. The types differ in their performance in the presence of faults and under influences from environmental conditions. In this part, the effects of electrical and electromechanical faults are considered (such faults are listed in Annex B). Additional requirements are provided in the other parts where faults generated by the particular sensing technology employed are considered. It is the responsibility of the machine supplier and/or the user to specify which type is required for a particular application.	Type 1 ESPE	P
	NOTE Requirements for a type 1 ESPE are not being considered at this time.		-
	A type 2 ESPE shall fulfil the fault detection requirements of 4.2.2.3.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	For a type 2 ESPE, in normal operation the output circuit of at least one output signal switching device shall go to the OFF-state when the sensing function is actuated, or when power is removed from the ESPE.		N/A
	A type 2 ESPE shall have a means of periodic test.		N/A
	A type 3 ESPE shall fulfil the fault detection requirements of 4.2.2.4.		N/A
	A type 4 ESPE shall fulfil the fault detection requirements of 4.2.2.5.		N/A
	For a type 3 ESPE and for a type 4 ESPE, in normal operation the output circuit of at least two output signal switching devices shall go to the OFF-state when the sensing function is actuated, or when power is removed from the ESPE.		N/A
	When a single safety-related data interface is used to perform the functions of the OSSD(s), then the data interface and associated safety-related communication interface shall meet the requirements of 4.2.4.4. In this case, a single safety-related data interface can substitute for two OSSDs in a type 3 or type 4 ESPE.		N/A
4.1.4	Types and required safety performance		-
	An ESPE shall meet a level of safety performance in accordance with IEC 62061 and/or ISO 13849-1, as stated in Table 1.	Not safety type ESPE	N/A
	NOTE The device-dependent probability of dangerous failure per hour (PFHD) values claimed for the control electronics are not restricted (for example, a manufacturer can claim a Type 2 has a PFHD lower than 10-6).		N/A
4.1.5	Required PLr or SIL and corresponding ESPE type		-
	In addition to the different levels of safety performance of the electrical parts of an ESPE control system, the potential risk reduction that can be provided by an ESPE is limited also by the systematic capabilities (for example, environmental influences, EMC, optical performance and detection principle). The limits are shown in Table 2.	Not safety type ESPE	N/A
	NOTE The intention of Table 2 is to limit the minimum type that should be employed for the risk reduction of a required safety function. For example: If a safety function requires SIL 2, then from Table 2, it can be seen that a Type 2 would not be sufficient.		N/A
4.2	Design requirements		-
4.2.1	Electrical supply		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The ESPE shall be designed to operate correctly with the conditions of the nominal supply as specified below, unless otherwise specified by the user:		P
	AC supplies Voltage: 0,85 to 1,1 of nominal voltage Frequency: 0,99 to 1,01 of nominal frequency (continuously) 0,98 to 1,02 of nominal frequency (short-time) Harmonics: Harmonic distortion not to exceed 10 % of the total RMS voltage between live conductors for the sum of the 2nd through to the 5th harmonic. An additional 2 % of the total RMS voltage between live conductors for the sum of the 6th through to the 30th harmonic is permissible.		N/A
	DC supplies From batteries Voltage: 0,85 to 1,15 of nominal voltage 0,7 to 1,2 of nominal voltage in the case of battery-operated vehicles From converting equipment Voltage: 0,9 to 1,1 of nominal voltage Ripple (peak-to-peak): Shall not exceed 0,05 of nominal voltage. For protection against electric shock, see 4.2.3.2.	From converting equipment Supply voltage: 9V DC-36V DC	P
	For protection against electrical interference, the power source should meet the requirements of IEC 61000-6-2.	Power adapter provided by user	N/A
4.2.2	Fault detection requirements		
4.2.2.1	General		
	The ESPE shall respond to the faults listed in Annex B, in accordance with 4.2.2.3 to 4.2.2.5 as appropriate. The faults listed in Annex B are not exclusive and, if necessary, additional faults shall be considered. For new components not mentioned in Annex B, a failure mode and effects analysis (FMEA, see IEC 60812) shall be carried out to establish the faults that are to be considered for those components.	Not safety type ESPE	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	From a lock-out condition, it shall not be possible for the ESPE to resume normal operation (for example, by interruption and restoration of the mains power supply or by reset) while the fault which initiated the lock-out condition is still present.	Not safety type ESPE	N/A
	At power on, the OSSD(s) shall not be signalled to the ON-state until it has been verified that there are no safety critical faults.	In the test, monitoring the output distance instead of the status of the OSSD	P
	The occurrence of single faults shall be considered by analysis and/or test (throughout the detection zone) with each of the following conditions:		P
	- environmental conditions specified in 4.3;		P
	- at the limits of alignment and/or adjustment.		P
4.2.2.2	Particular requirements for a type 1 ESPE		-
	NOTE Particular requirements for a type 1 ESPE are not under consideration at this time.	Type 1 ESPE	P
4.2.2.3	Particular requirements for a type 2 ESPE		-
	A type 2 ESPE shall have a means of periodic test to reveal a failure to danger (for example loss of detection capability, response time exceeding that specified). The test shall also be performed at power-on of the ESPE before going to the ON-state and at each reset.		N/A
	Depending on the application, it can be necessary to perform the periodic test more often to achieve a desired safety performance. Generic functional safety standards give requirements about how often periodic tests have to be applied to fulfil the requirements for a certain safety performance.		N/A
	NOTE The periodic test can be initiated by external or internal means.		-
	A single fault resulting in the loss of detection capability or the increase in response time beyond the specified time or preventing one or more of the OSSDs going to the OFF-state, shall result in a lock-out condition as a result of the next periodic test.		N/A
	Where the periodic test is intended to be initiated by an external (for example machine) safety-related control system, the ESPE shall be provided with suitable input facilities (for example terminals).		N/A
	The duration of the periodic test shall be such that the intended safety function is not impaired.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	If the type 2 ESPE is intended for use as a trip device (for example when used as a perimeter guard), and the duration of the periodic test is greater than 150 ms, it is possible for a person to pass through the detection zone without being detected. In this case a restart interlock should be included.		N/A
	An ESPE with only one OSSD shall have a minimum of one SSD (see Clause A.4).		N/A
4.2.2.4	Particular requirements for a type 3 ESPE		-
	A single fault in the sensing device resulting in a complete loss of the stated ESPE detection capability shall cause the ESPE to go to a lock-out condition within the specified response time.		N/A
	A single fault resulting in a deterioration of the stated ESPE detection capability shall cause the ESPE to go to a lock-out condition within a time period of 5 s following the occurrence of that fault.		N/A
	EXAMPLE Drift and aging of components can cause a fault resulting in deterioration of the detection capability.		-
	A single fault resulting in an increase in response time beyond the specified value or preventing at least one OSSD going to the OFF-state shall cause the ESPE to go to a lock-out condition within the response time, or immediately upon any of the following demand events where fault detection requires a change of state:		N/A
	- on actuation of the sensing function;		N/A
	- on power off/on;		N/A
	- on reset of the start interlock or the restart interlock, if available (see Clauses A.5 and A.6);		N/A
	- on the application of an external test signal, if available.		N/A
	In cases where a single fault which in itself does not cause a failure to danger is not detected, the occurrence of one additional fault shall not cause a failure to danger. For verification of this requirement, see 5.3.4.		N/A
4.2.2.5	Particular requirements for a type 4 ESPE		-
	A single fault resulting in a loss of detection capability shall cause the ESPE to go to a lockout condition within the response time.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	A single fault resulting in an increase in response time beyond the specified value or a single fault preventing one or more than one OSSD going to the OFF-state, shall cause the ESPE to go to a lock-out condition within the response time, or immediately upon any of the following demand events where fault detection requires a change of state:		N/A
	- on actuation of the sensing function;		N/A
	- on power off/on;		N/A
	- on reset of the start or restart interlock, if available (see Clauses A.5 and A.6).		N/A
	In cases where a single fault which in itself does not cause a failure to danger is not detected, the occurrence of further faults shall not cause a failure to danger. For verification of this requirement, see 5.3.5.		N/A
	NOTE 1 Design measures for a type 4 ESPE may include: - single-channel technique with dynamic fault detection measures; or - single-channel technique with an internally generated automatic check, performed frequently so that the automatic check interval for fault detection is included in the safety device response time; and - multiple channel techniques such that any disparity between channels results in a lock-out condition.		-
	NOTE 2 For additional requirements for integrated circuits, complex or programmable, see 4.2.10.		-
4.2.3	Electrical equipment of the ESPE		-
4.2.3.1	General		-
	The electrical equipment (components) of the ESPE shall:		-
	- conform to appropriate IEC standards where they exist;	Approved components used	P
	- be suitable for the intended use; and	See above	P
	- be operated within their specified ratings.	See above	P
4.2.3.2	Protection against electric shock		-
	Protection against electric shock shall be provided in accordance with Clause 6 of IEC 60204‑1:2016.	PELV circuit	P
4.2.3.3	Protection of electrical equipment		-
	Overcurrent protection shall be provided in accordance with 7.2.1, 7.2.3, 7.2.7, 7.2.8, and 7.2.9 of IEC 60204-1:2016.	Considered	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE Information may need to be given to the user of the ESPE as to the maximum rating of fuses, or setting of an overcurrent protective device for the circuit(s) connected to the OSSD(s) output connection points.		-
4.2.3.4	Pollution degree		-
	The electrical equipment shall be suitable for pollution degree 2 (as defined in 6.1.3.2 of IEC 60947-1:2007).		P
4.2.3.5	Clearance, creepage distances and isolating distances		-
	The electrical equipment shall be designed and constructed in accordance with 7.1.4 of IEC 60947-1:2007/AMD2:2014.		P
4.2.3.6	Wiring		-
	The electrical equipment shall be wired in accordance with IEC 60204-1:2016.		P
4.2.3.7	Fire and flame protection		-
	Fire and flame protection shall be provided in accordance with relevant standards		P
	NOTE IEC 60204-1, IEC 60947-1 and IEC 61010-1 are examples of relevant standards that can provide guidance.		-
4.2.4	Output signal switching devices (OSSDs)		-
4.2.4.1	General		-
	Separate output connection points (terminals) shall be provided for each OSSD.	No OSSDs	N/A
	The OSSDs should be so rated that their loads can be switched without the use of arc suppression devices fitted across the loads.		N/A
	The output circuit of the OSSDs should be adequately protected to prevent failure to danger, for example welded contacts under overcurrent conditions (see 7.2.9 of IEC 60204-1:2016).		N/A
	Measures should be provided to minimize the possibility of failure to danger from common cause failures.	Not safety type	N/A
	Some functions of the machine safety-related control system may be performed by the ESPE, for example the OSSD may perform the function of an FSD.	Not safety type	N/A
	Both a type 3 ESPE and a type 4 ESPE shall incorporate a minimum of two independently operated OSSDs.	Type 1 ESPE	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	A reference to an OSSD action (for example, go to the OFF-state) will also mean a corresponding action of a safety-related data interface. A single safety-related data interface can meet the requirements of having two OSSDs.	Not safety type	N/A
	If relay OSSDs are provided, the state (i.e. position) of the contacts shall be monitored. This can be achieved by monitoring the state of an auxiliary contact(s) on relays with mechanically linked (positively guided) contacts. The mechanical link ensures that the monitored contact follows the change of state of the OSSD contact(s).	No relay OSSDs	N/A
	Special design and constructional measures shall be used to ensure that the make (normallyopen) contact(s) and the break (normally-closed) contact(s) cannot be in the closed position simultaneously.	Not safety type	N/A
	NOTE 1 The mechanical link ensures that the monitored contact follows the change of state of the OSSD contact(s).		N/A
	NOTE 2 It is important that relay drop out voltage and the separation distance between the contacts are maintained at a proper level over the entire stated life of the relay.		N/A
4.2.4.3	Solid state OSSDs		-
	Solid state OSSD outputs may be either current sourcing or current sinking types. When current sourcing outputs are provided, they shall meet the requirements of this subclause 4.2.4.3.	No this device	N/A
	NOTE 1 Requirements for current sinking outputs which can be required for certain applications are not defined in this document. When current sinking outputs are used, a short-circuit to the reference potential or an open circuit can be interpreted by the inputs and loads as the ON-state (see IEC 60204-1:2016, 9.4.3.1).		N/A
	NOTE 2 The values given in the table below meet the requirements of IEC 61131-2:2017 (see 6.4.4 of IEC 61131‑2:2017), for a nominal rated supply voltage of 24 V DC. When other supply voltages are used, this document may be used as a guide. IEC 61131-2:2017 can be referred to for additional information.		N/A
	The output(s) shall be protected against the effects of overvoltage, overcurrent and short circuit.		N/A
	The maximum leakage current shall not exceed 2 mA.		N/A
	NOTE 3 It is possible that a leakage current greater than 2 mA can lead to a failure to danger.		N/A
	When there is more than one OSSD, short circuits between the outputs of the OSSDs shall be detected.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The supplier of the ESPE shall provide the following information in the accompanying documents:		N/A
	- nominal and maximum output current in the ON-state for resistive and inductive loads;		N/A
	- maximum OFF-state voltage;		N/A
	- maximum output current in OFF-state (leakage current);		N/A
	- maximum capacitive load;		N/A
	- maximum resistance of the connection(s) between the OSSD(s) and the load(s).		N/A
4.2.4.4	Safety-related data interface and safety-related communication interface		-
	When the sensing device is actuated during normal operation, the ESPE shall respond by sending information indicating the status of the sensing device or ESPE through a safetyrelated data interface. The status information is converted to a data telegram by a safetyrelated communication interface.	Not safety type	N/A
	The safety-related data interface shall have the same protection against faults as is appropriate for the type of ESPE.		N/A
	Depending on the ESPE design, the safety-related communication interface can either be external in a separate enclosure (Figure 1a) or it can be integrated in the same enclosure of the ESPE (Figure 1b).		N/A
	When the safety-related communication interface is integrated in the ESPE, the entire ESPE shall meet the relevant requirements of IEC 62061 or IEC 61508 (all parts).		N/A
	NOTE Because of the specific technology of communication interfaces, different standards other than this document apply. To avoid overlapping with other standards, functional requirements for the safety-related communication interface are not defined in this document.		N/A
4.2.5	Indicator lights and displays		-
	Devices shall be provided by the ESPE supplier to:		P
	a) indicate the actuation of the sensing device. Neither the time from the actuation of the sensing device to the indicator achieving 50 % of its final brightness (luminescence), nor the time from the de-actuation of the sensing device to the indicator brightness decaying to 50 % of its initial brightness, shall exceed 100 ms;	Indicator lights used	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	b) indicate the output status of an OSSD. The ON-state shall be represented by a green indicator, the OFF-state by a red indicator. When two or more OSSDs are intended to operate in co-ordination, a single set of indicators may be shared.	No OSSDs	N/A
	When there are two or more indicators of the same colour, the function of each indicator shall be unambiguously marked.	Two indicators with different color	N/A
	NOTE For some modes of operation, the same set of indicators for a) can also be used for b). A bi-colour indicator could be used.		-
	The indicators are intended for the machine operator. Therefore they shall be capable of being located near the detection zone and visible when the equipment is installed. They can be integrated in the sensor elements or as an external equipment installed near by the detection zone.		P
4.2.6	Adjustment means		-
	All adjustment means shall be so designed that a failure to danger is not possible at any point in the range of adjustment. A failure in the adjustment means shall not cause an unintended change to the configuration of the ESPE.		P
4.2.7	Disconnection of electrical assemblies		-
	When means are provided to permit disconnection of any subsystem, part of a subsystem or any plug-in component, such disconnection shall result in at least one OSSD going to the OFF-state, in accordance with 4.2.2. This requirement includes disconnections both within a single enclosure and/or between separate enclosures (for example a master/slave sensor configuration).	No this situation	N/A
4.2.8	Non-electrical components		-
	Non-electrical components shall be suitable for the intended use.	Considered	P
4.2.9	Common cause failures		-
	The design should be such as to minimize the possibility of a failure to danger from common cause failures arising from:		P
	- environmental influences;		P
	- multichannel systems using a common substrate;	Not multichannel system	N/A
	- short circuits between channels of multichannel systems.	Not multichannel system	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE 1 Common cause failures can also result from the use of components degraded by mishandling, faulty manufacture, etc.		-
	NOTE 2 Common cause failures are treated as a single failure.		-
	Where a common semi-conductor substrate is used for more than one channel of a multichannel system, the special architecture requirements for on-chip redundancy of IEC 61508-2 apply.	Not multichannel system	N/A
	NOTE 3 In IEC 61508-2:2010, this is Annex E.		-
4.2.10	Programmable or complex integrated circuits		-
	Where programmable or complex integrated circuits are used in a type 4 ESPE, the safetyrelated performance shall be maintained by at least two independent controlling/monitoring channels. This requirement shall be verified in accordance with 5.5.	Type 1 ESPE	N/A
4.2.11	Software, programming, functional design of integrated circuits		-
4.2.11.1	General		-
	Where an ESPE implements its safety-related performance by any of the following means, the additional requirements of 4.2.11.2 apply:	Not safety type ESPE	N/A
	a) a software program(s) executed during operation;		N/A
	b) a programmed device(s), the functions of which were set by a process subsequent to its original manufacture, for example programmable logic devices and read only memories;		N/A
	c) a device(s) manufactured to a specific user functional specification, for example ASIC, mask programmed microprocessor, ROM.		N/A
	Conformance to these requirements shall be validated in accordance with 5.5.		N/A
4.2.11.2	Requirements		-
	The software, device program and the device functional design shall be developed in accordance with IEC 61508-3 for the appropriate SIL or in accordance with ISO 13849-1 for the appropriate PL.		N/A
4.2.12	Integrity of the ESPE detection capability		-
	The design of the ESPE shall ensure that the detection capability is not decreased below the limits specified by the supplier and in a subsequent part of IEC 61496 that provides particular requirements based on the sensing technology.	EN IEC 61496-3	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
4.2.13	Test piece		-
	A test piece shall be defined as part of the ESPE design for use in verifying the detection capability. The test piece shall be defined based on the sensing technology and test requirements as defined in Clause 5 of the relevant parts of IEC 61496.		P
4.3	Environmental requirements		-
4.3.1	Ambient air temperature range and humidity		-
	The ESPE shall comply with the requirements of this standard when subjected to ambient temperature variations from 0 °C to 50 °C. Where it is intended for use outside this range, the supplier shall specify the temperature range over which the system will continue normal operation. Compliance with this requirement shall be verified by the tests specified in 5.4.2.2 at a non-condensing humidity of 95 % for temperatures between 20 °C and the highest ambient temperature in accordance with 5.4.2.1.	Tested	P
	When the supplier states that the ESPE can be used in an environment where condensing is possible, the ESPE shall not fail to danger when subjected to a rapid change of temperature and humidity leading to condensation on the ESPE. This requirement is verified by the condensing test of 5.4.2.3.	Tested	P
4.3.2	Electrical disturbances		-
4.3.2.1	Supply voltage variations		-
	The ESPE shall not fail to danger when the external supply voltage is reduced steadily and continuously from the nominal voltage to zero voltage, over a period of 10 s to 20 s, and then increased in a similar manner from zero voltage to the nominal voltage.	Tested	P
	The ESPE shall not fail to danger when each internally derived supply voltage, in turn, is varied steadily and continuously over a period of 10 s to 20 s, from nominal voltage to zero voltage, and then increased in a similar manner from zero voltage to nominal voltage.	No internal power supply	N/A
4.3.2.2	External supply voltage dips and interruptions		-
	For AC powered devices, the requirements of Table 3 apply for supply voltage dips and interruptions in accordance with IEC 61000-4-11:2020.	DC powered device	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	For devices supplied by external DC networks, the requirements of Table 4 apply for supply voltage dips and interruptions in accordance with IEC 61000-4-29:2000.	Tested	P
	When a DC powered ESPE is designed to be supplied from a specific type of power supply(s) (for example, supplied direct from a safety-related communication interface), the supply voltage dips and interruptions in 4.3.2.2 may be applied to the primary input of the specified power supply instead of direct to the ESPE.	No this situation	N/A
4.3.2.3	Burst		-
4.3.2.3.1	General requirements		-
	The ESPE shall continue in normal operation when subjected to burst in accordance with IEC 61000-4-4:2012:	Tested	P
4.3.2.3.2	Additional requirements		-
	A type 3 or a type 4 ESPE shall not fail to danger when subjected to burst in accordance with IEC 61000-4-4:2012:	Type 1 ESPE	N/A
4.3.2.4	Surge		-
4.3.2.4.1	General requirements		-
	The ESPE shall continue in normal operation when subjected to surge in accordance with IEC 61000-4-5:2014 and IEC 61000-4-5:2014/AMD1:2017:	Tested	P
4.3.2.4.2	Additional requirements		-
	A type 3 or a type 4 ESPE shall not fail to danger when subjected to surge in accordance with IEC 61000-4-5:2014 and IEC 61000-4-5:2014/AMD1:2017:	Type 1 ESPE	N/A
4.3.2.5	Electromagnetic field		-
4.3.2.5.1	General requirements		-
	The ESPE shall continue in normal operation when subjected to an electromagnetic field in accordance with IEC 61000-4-3:2006, IEC 61000-4-3:2006/AMD1:2007 and IEC 61000 - 4 - 3:2006/AMD2:2010:	Tested	P
4.3.2.5.2	Additional requirements		-
	A type 3 or a type 4 ESPE shall not fail to danger when subjected to an electromagnetic field in accordance with IEC 61000-4-3:2006, IEC 61000-4-3:2006/AMD1:2007 and IEC 61000 - 4 - 3:2006/AMD2:2010:	Type 1 ESPE	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The test of 1,4 GHz to 6,0 GHz is required only at the frequencies or frequency ranges defined in IEC 61000-6-7:2014, Table 7 - "General frequency ranges for mobile transmitters and ISM equipment for radiated tests".		N/A
4.3.2.6	Conducted disturbances induced by radio-frequency fields		-
4.3.2.6.1	General requirements		-
	The ESPE shall continue in normal operation when subjected to conducted radio-frequency disturbances in accordance with IEC 61000-4-6:2013:	Tested	P
4.3.2.6.2	Additional requirements		-
	A type 3 or a type 4 ESPE shall not fail to danger when subjected to conducted radiofrequency disturbances in accordance with IEC 61000-4-6:2013:	Type 1 ESPE	N/A
4.3.2.7	Electrostatic discharge		-
4.3.2.7.1	General requirements		-
	The ESPE shall continue in normal operation when subjected to an electrostatic discharge in accordance with IEC 61000-4-2:2008:	Tested	P
4.3.2.7.2	Additional requirements		-
	A type 3 or a type 4 ESPE shall not fail to danger when subjected to an electrostatic discharge in accordance with IEC 61000-4-2:2008:	Type 1 ESPE	N/A
4.3.3	Mechanical environment		-
4.3.3.1	General		-
	The supplier shall specify the possible application(s) of the ESPE by selecting at least one of the following:		P
	a) If the supplier specifies the ESPE for use in stationary applications, the supplier shall choose an appropriate 3M class in accordance with IEC TR 60721-4-3. Preference is given to classes 3M4 and 3M7, see Tables 5 and 8. If the preferred classes are not representative for the application the supplier may specify alternative conditions.	Stationary application considered	P
	b) If the supplier specifies the ESPE for use on ground vehicles, the supplier shall choose an appropriate 5M class in accordance with IEC 60721-3-5. If available classes 5M are not representative for the application the supplier may specify alternative conditions.	Ground vehicle application considered	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE 1 Different parts of an ESPE, e.g. sensing device and control device, can be assigned to different classes depending on the manufacturer's specification of their application.		-
	NOTE 2 Specification of a 5M class does not mean that the impact from foreign bodies, stones have to be tested in accordance with IEC 60068-2-75 in general. Other parts of the IEC 61496 series can specify such a test.		-
4.3.3.2	Vibration		-
	The ESPE shall pass the vibration tests of 5.4.4.1.	Tested	P
4.3.3.3	Shock		-
	The ESPE shall pass the shock tests of 5.4.4.2.	Tested	P
4.3.4	Enclosures		
	The ESPE shall have its own enclosure(s).	Tested	P
	All enclosures of the ESPE, including those mounted remotely, shall provide a degree of protection of at least IP54 (see IEC 60529), when mounted as specified by the supplier. However, when mounted in a machine controlgear enclosure having a degree of protection of at least IP54, the ESPE enclosure shall have a degree of protection of at least IP20.	IP67 reached	P
	NOTE Protection against mechanical damage can be achieved by: - a suitable location; - the use of suitable materials and form of construction providing adequate strength; or - the use of a protective barrier.		-
	The method of cable entry for incoming cables shall not impair the degree of protection.	Considered	P
	Sealing compounds which adhere to the two surfaces being joined, such that the environmental protection is degraded when the joint is separated, shall not be used to seal covers which might be removed for service access.	No this situation	N/A
	Enclosures shall be free from sharp edges or corners capable of causing damage to cable insulation. Compliance shall be checked by inspection.	Smooth surfaces	P
	Enclosures shall provide adequate access to enable any necessary adjustments and maintenance work to be carried out safely and effectively. The covers enabling such access shall have captive fasteners.	Considered	P
4.3.5	Light interference		-
	When an ESPE uses technology that can be influenced by ambient light conditions, that ESPE shall meet the requirements of 4.3.5.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Where other parts of IEC 61496 refer to "light interference", the requirements and tests of that part apply.	EN IEC 61496-3	P
	The ESPE shall continue in normal operation when subjected to	Tested	P
	- incandescent light;		P
	- fluorescent light operated with high-frequency electronic power supply.		P
	The ESPE shall not fail to danger when subjected to		-
	- incandescent light (simulated daylight using a quartz lamp);		P
	- fluorescent light operated with high-frequency electronic power supply;		P
	- flashing beacons;		P
	- collimated laser beam;		P
	- for a type 3 and type 4 ESPE, radiation from an emitting assembly (or element) of identical design. Combinations of technical measures and configuration procedures in accordance with the information for use provided by the supplier shall be tested.	Type 1 ESPE	N/A
	NOTE 1 Radiation from an emitting assembly of identical design is considered to be a worse case example of interference from a LED lamp.		--
	NOTE 2 For type 2 ESPEs the risk of failure to danger from an emitting element of identical design can be reduced by installation measures provided by the supplier.		--
	These requirements shall be met when the ESPE conforms to the tests in 5.4.6.		P
	No requirements are given for immunity to other extraneous light sources which may cause abnormal operation or failure to danger. A requirement for the supplier to inform the user of potential problems is given in Clause 7.		P
5	**Testing**		-
5.1	General		-
5.1.1	Type tests		-
5.1.1.1	Test samples		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	In so far as it is practicable, all parts of an ESPE shall be tested together. Where this is not practicable, parts of the ESPE may be tested separately. Examples of such situations include integrated ESPEs (ESPEs integrated in, and normally not separable from machinery) in the case of environmental tests. In such cases:	Tested together	P
	- any input signals necessary for the operation of the ESPE shall be simulated;		P
	- these exceptions and any omissions of tests shall be stated in the test report.		P
	Where a particular test would be destructive and identical results could be obtained by testing part of the ESPE in isolation, a sample of that part may be used instead of the whole equipment sample for the purpose of obtaining the results of the test.		P
	Where the ESPE is designed for operation at a number of different supply voltages (for example for differing applications), more than one sample may be required.	No this situation	N/A
	When the ESPE is designed to be supplied from an external dedicated power supply, the ESPE shall undergo testing with the specified dedicated power supply (see 6.2).		P
	The tests of 5.4.5 shall be performed after the tests of 5.4.2 and 5.4.4 have been completed. The same test sample shall be used in all three tests.		P
5.1.1.2	Operating condition		-
	Unless otherwise stated in the test procedure, the tests shall be carried out with the test sample operating within the conditions specified in the accompanying documents.		P
	For the purposes of electrical disturbance immunity tests, the equipment shall be in as near its final operating configuration as is possible (i.e. with all peripheral devices and covers attached, connected to the power supply and, where applicable, connected to the external protective conductor and/or the external functional bonding conductor (see IEC 60204-1:2016)).		P
	When several mounting positions are specified, the least favourable mounting position shall be used.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Where a safety-related data interface is used in place of an OSSD(s), the ESPE shall be connected to a communication system in accordance with the supplier's instructions that has a means of monitoring the ESPE status.		N/A
	As a result of the analysis of the design and the integrity of the detection capability of the ESPE, a test plan shall be established considering the test conditions and parameters outlined in this document. In the following tests, it shall be verified that when the OSSDs go to the OFF-state, they remain in the OFF-state while the test piece is present in the detection zone.	No OSSD	N/A
	The minimum test conditions shall be as specified in this document or by the supplier, whichever is more stringent.		P
	Tests can be omitted if an analysis can demonstrate that	No this situation	N/A
	- limited intended use in application justifies omission of test (e.g. only for top mounting, for use only in non-condensing environment), or		N/A
	- a less stringent test is replaced by a more stringent test (e.g. test of equal detection capability at minimum working range versus maximum working range), or		N/A
	- detection capability is not affected by the environmental condition (e.g. condensation).		N/A
	The analysis shall be documented in the test plan.		N/A
	For some ESPEs, the tests specified in 5.4.2, 5.4.3 and 5.4.4 can have limitations with respect to test equipment and test setups. Any deviation from the test requirements given in these clauses shall be accompanied by a full description and technical justification in the test report.		P
5.1.1.3	Simulated intrusion into detection zone		-
	In the following tests, introduction of the test piece (as defined in the relevant part of IEC 61496) into the detection zone may be simulated if the method can be shown to be equivalent.		P
5.1.2	Test conditions		-
5.1.2.1	Test environment		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Except where otherwise specified in 5.4, the tests shall be carried out with the ESPE operated under the following conditions:	Tested in following conditions	P
	- rated voltage (or a voltage within the rated voltage range);		P
	- rated frequency (or a frequency within the rated frequency range);		P
	- ambient temperature: 20 °C ± 5 °C;		P
	- relative humidity: 25 % to 75 %;		P
	- barometric pressure: 86 kPa to 106 kPa.		P
	NOTE Values stated in the marking and in the accompanying documents are considered as rated values.		-
5.1.2.2	Measurement accuracy		-
	The errors of measurement shall not exceed:		P
	- for measurement of ESPE response time: ±1 ms;		P
	- for temperature measurements: ±3 °C;		P
	- for electrical measurements: ±1 %, where technically possible and/or appropriate;		P
	- for relative humidity (RH) measurements: ±3 % RH;		P
	- for linear measurements: ±1 mm or ±1 %, whichever is the greater.		P
	- for angular measurement: ±0,1°;		P
	- for light intensity measurement: ±10 %.		P
	All measurements shall be made after stable temperature conditions have been achieved. This is considered to be fulfilled when the rate of temperature rise or fall is less than 2 K/h.		P
	The ESPE and the safety-related communication interface shall be tested together (see Figure 2). Due to the fact that the safety-related communication interface does not show a static output signal, it is necessary to use a data receiver. The test setup consists of the equipment under test and a data receiver (for example, a PLC or monitoring device) which indicates the status of the sensing device or ESPE.	Not safety type	N/A
	When testing for susceptibility to electrical disturbances, an appropriate test adapter which isolates the ESPE under test from the communication bus may be required.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.1.3	Test results		-
	The results of all tests and analyses shall be documented. The test results shall be arranged in a form that shows the details of each individual test and its effects. Details of any special test procedures shall be included in the test report.		P
5.2	Functional tests		-
5.2.1	Sensing function		-
	The sensing function and the integrity of the detection capability and zone (for example size, shape and location) of the ESPE shall be verified as specified in the relevant part of IEC 61496.	EN IEC 61496-3	P
5.2.2	Response time		-
	The response time shall be verified by systematic analysis and test.	Not safety type ESPE	N/A
	The response time may be determined by electrical simulation of an actuation, provided that it includes the maximum time between the event triggering the actuation of the sensing device and the actuation.		N/A
	Additional requirements for the measurement of the response time of the ESPE may be given in the relevant part of IEC 61496.		N/A
5.2.3	Limited functional tests		-
5.2.3.1	General		-
	The following limited functional tests A, B and C shall be performed to verify that under normal environmental conditions the ESPE shall continue normal operation and that, under abnormal environmental conditions or under fault conditions, the ESPE shall not fail to danger.		P
	When an ESPE is fitted with a restart interlock function, that function shall be bypassed or not selected during the performance of the tests. The restart interlock function shall be tested separately (see Annex A).		N/A
	When using a safety-related communication interface, in the following limited functional tests, OSSDs going to the ON-state or OFF-state are replaced by a safety-related message (for example, a data telegram) indicating the corresponding status of the sensing device or ESPE.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE In some situations, it can be necessary to simulate the actuation of the sensor by some means other than introducing a test piece.		-
5.2.3.2	Limited functional test A (A test)		-
	With no intrusion in the detection zone, it shall be verified, for a period of at least 5 s, unless otherwise specified, that the OSSD(s) shall be in the ON-state and shall not go to OFF-state.	Monitors the obstacle distance output from radar in place of OSSD status	P
5.2.3.3	Limited functional test B (B test)		-
	With no intrusion in the detection zone, it shall be verified, for a period of at least 5 s, unless otherwise specified, that the OSSD(s) shall be in the ON-state and shall not go to the OFFstate.	Monitors the obstacle distance output from radar in place of OSSD status	P
	The test piece shall be inserted into the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state. It shall be verified that, for a period of at least 5 s unless otherwise specified, the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.	Monitors the obstacle distance output from radar in place of OSSD status	P
	The test piece shall be removed from the detection zone or the sample shall be otherwise deactivated. The OSSD(s) shall respond by going from the OFF-state to the ON-state. It shall be verified that, for a period of at least 5 s unless otherwise specified, the OSSD(s) remain(s) in the ON-state with the test piece not being present in the detection zone.	Monitors the obstacle distance output from radar in place of OSSD status	P
	It can be necessary to repeat the above tests continuously depending on test requirements.		P
5.2.3.4	Limited functional test C (C test)		-
	This test is the same as the limited functional test B but, where the OSSD(s) should be in the ON-state, the OSSD(s) are allowed to be in the OFF-state. No failure to danger shall occur. At the end of each of the relevant tests in 5.4, the ESPE shall continue to operate normally or shall resume normal operation following recovery from a lock-out condition.	Monitors the obstacle distance output from radar in place of OSSD status	P
	If the ESPE cannot resume normal operation due to a permanent component failure, it is acceptable if it is verified that the OSSD(s) remains in the OFF-state.	Monitors the obstacle distance output from radar in place of OSSD status	P
5.2.4	Periodic test		-
	For a type 2 ESPE, the requirements of 4.2.2.3 shall be verified by analysis and measurement.	Type 1 ESPE	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.2.5	Indicator lights and displays		-
	The functions and colours of indicator lights and displays shall be verified in accordance with the requirements of 4.2.5 by applying a B test.		P
5.2.6	Means of adjustment		-
	The requirements of 4.1.1 and 4.1.2 shall be verified by inspection. The requirements of 4.2.6 shall be verified by inspection and by carrying out C tests as necessary.		P
5.2.7	Rating of components		-
	The operation of each component within its specified ratings, throughout the entire operating range of the ESPE, shall be verified by analysis and/or inspection.		P
5.2.8	Output signal switching devices (OSSD)		-
5.2.8.1	General		-
	The provision of separate output connection points (terminals) for each OSSD shall be verified by inspection.	No OSSD	N/A
	When two OSSDs are provided, verify by inspection or test that the OSSDs are operated independently.		N/A
	Verify by inspection that the OSSD(s) are protected by current limiting devices or that information for installation of current limiting devices is provided in information for use.		N/A
	It shall be verified that foreseeable faults will not cause the OSSD(s) to go to or remain in the ON-state when they should be in the OFF-state. All tests shall be carried out with the maximum inductive load and the maximum length of the connection cable specified by the manufacturer.		N/A
	Foreseeable faults include:		N/A
	- short circuit of the OSSD to supply voltage;		N/A
	- short circuit of the OSSD to ground;		N/A
	- short circuit between the OSSDs;		N/A
	- open circuit at the power supply return cable;		N/A
	- open circuit of the functional bonding conductor;		N/A
	- open circuit of the screen of a screened cable;		N/A
	- incorrect wiring.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.2.8.2	Relay OSSDs		-
	Verify by inspection or tests that the relays meet the requirements of 4.2.4.2.		N/A
5.2.8.3	Solid state OSSDs		-
	The output voltage and current levels specified in 4.2.4.3 shall be verified.		N/A
5.2.8.4	Safety-related data interface and safety-related communication interface	Not safety type ESPE	-
	It shall be verified by test that disconnection of components does not lead to a failure to danger.		N/A
	Electrical tests specified in 5.2.8.1 (shorts, opens, improper loading) which would be applied to OSSD(s) may be excluded if not applicable.		N/A
	The safety integrity of an integrated communication interface shall be verified by tests, systematic analysis and by reference to data sheets and test reports in accordance with the requirements of 4.2.4.4.		N/A
5.3	Performance testing under fault conditions		-
5.3.1	General		-
	Tests for the effects of single faults selected in accordance with 4.2.2 shall be carried out on all the relevant components of the ESPE. If further faults occur as a result of the first single fault, the first and all consequent faults shall be considered as a single fault.	Not safety type	N/A
	A fault catalogue including all components shall be prepared recording the results of the fault considerations listed in Annex B. In order to reduce unnecessary testing in accordance with 5.3.3, 5.3.4 and 5.3.5 where the results of a single fault or combination of faults can be predicted theoretically, an analysis statement shall be included as part of the test results statement. That statement shall be validated in accordance with 5.5.4. In such cases, only selected (sample) tests need to be carried out to confirm such analysis statements.		N/A
	NOTE 1 Typical methods used for the fault assessment include fault mode and effect analysis (FEMA) in accordance with IEC 60812 and fault tree analysis (FTA) in accordance with IEC 61025		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	NOTE 2 In the case of complex circuit structures or components (for example microprocessor, complete redundancy) the review of faults is generally carried out on the structural level. See Clause B.2 for exclusion of short circuits on assembled circuit boards and for exclusion of short circuits between adjacent terminals for external connection.		-
5.3.2	Type 1 ESPE		-
	NOTE Requirements for a type 1 ESPE are not being considered at this time.	Type 1 ESPE	N/A
5.3.3	Type 2 ESPE		-
	The ESPE shall be subjected to single faults to establish that a fault leading to a hazardous condition (for example loss of detection capability or increased response time) is detected by the periodic test function and results in a lock-out condition in accordance with 4.2.2.3.		N/A
5.3.4	Type 3 ESPE		-
	The ESPE shall be subjected to single faults to establish that the fault is detected by the ESPE going to a lock-out condition and that no failure to danger occurs, in accordance with 4.2.2.4.		N/A
	When a single fault is not detected and the analysis specified in 5.3.1 cannot be carried out, the tests for the ESPE going to a lock-out condition and no failure to danger occurring shall be continued with that fault applied first and all other faults added and removed in turn. Tests shall be carried out for all undetected single faults.		N/A
	Testing for accumulation of more than two faults need not to be carried out.		
5.3.5	Type 4 ESPE		-
	The ESPE shall be subjected to single faults to establish that the fault is detected by the ESPE going to a lock-out condition and that no failure to danger occurs, in accordance with 4.2.2.5.		N/A
	When a single fault is not detected and the analysis specified in 5.3.1 cannot be carried out, the tests for the ESPE going to a lock-out condition and no failure to danger occurring shall be continued with that fault applied first and all other faults added and removed in turn. Tests shall be carried out for all undetected single faults.		N/A
	Testing for the accumulation of more than two faults need not to be carried out.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.4	Environmental tests		-
5.4.1	Rated supply voltage		-
	The design measures specified in 4.2.1 shall be verified by inspection.		P
	The ESPE shall be subjected to the following sequence of tests using the relevant values specified in 4.2.1.	Tested	P
	a) The ESPE shall be supplied with the lowest rated supply voltage. A B test shall be carried out.		P
	b) The supply voltage shall be increased, within a period of 10 s to 20 s, to the highest rated voltage, during which time an A test shall be performed.		P
	c) After the highest test supply voltage has been reached, a B test shall be carried out.		P
	The requirements for frequency variation and for harmonic distortion shall be verified either by testing or by using analytical methods.	Tested	P
5.4.2	Ambient temperature variation and humidity		-
5.4.2.1	General		-
	The highest ambient temperature in the tests below shall be as specified in the marking and/or the accompanying documents, but shall not be lower than +50 °C. The lowest ambient temperature in the tests below shall be as specified in the marking and/or the accompanying documents, but shall not be higher than 0 °C.	Tested	P
5.4.2.2	Temperature variation and humidity test sequence		-
	The ESPE shall be subjected to the following sequence of tests.	Tested	P
	a) With the ESPE operating under the conditions specified in 5.1.2.1, an A test shall be performed with a duration of at least 2 h. At the end of that time, a B test shall be carried out.		P
	b) The ambient temperature shall be increased by not more than 0,3 °C per minute up to the highest ambient temperature, during which time an A test shall be performed.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	c) An A test shall be performed with a duration of at least 2 h at the highest ambient temperature. During that time, the humidity shall be increased to 95 % and held at that value for at least 1 h. Following the A test, a B test shall be carried out.		P
	d) The ambient temperature shall be decreased by not more than 0,3 °C per minute, whilst maintaining the humidity at 95 %, until a temperature of 20 °C is reached, during which time an A test shall be performed.		P
	e) The ambient temperature shall be decreased by not more than 0,3 °C per minute, without condensation occurring, until the lowest ambient temperature is reached, during which time an A test shall be performed.		P
	f) An A test shall be performed with a duration of at least 2 h at the lowest ambient temperature. At the end of that time, a B test shall be carried out.		P
	g) The ambient temperature shall then be increased by not more than 0,3 °C per minute to the value specified in 5.1.2.1, during which time an A test shall be performed.		P
	h) An A test shall be performed with a duration of at least 2 h at the temperature specified in 5.1.2.1. At the end of that time a B test shall be carried out.		P
5.4.2.3	Condensing test procedure		-
	The ESPE shall be subjected to the following condensing test.	Tested	P
	a) The ESPE shall be supplied with its rated voltage and stored in a test chamber at an ambient temperature of 5 °C for 1 h.		P
	b) The ambient temperature and the humidity shall be changed within a time period of up to 2 min to a temperature of (25 ± 5) °C and a relative humidity of (70 ± 5) %.		P
	c) A C test shall be performed with a duration of 10 min using a test piece.		P
	d) If a restart interlock is available, it shall not be operational during the C test.		P
5.4.3	Effects of electrical disturbances		-
5.4.3.1	Supply voltage variations		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The external supply voltage and each internally derived supply voltage shall in turn be varied in accordance with 4.3.2.1. During each test, a C test(s) shall be carried out as necessary to confirm that no failure to danger occurs at reduced voltage values.	Tested	P
5.4.3.2	Supply voltage dips and interruptions		-
	Each applicable test specified in 4.3.2.2 shall be performed at least 10 times in the ON state of the ESPE and 10 times in the OFF state. The OFF state shall be achieved by inserting a test piece into the detection zone.	Tested	P
	During tests 1 and 2, the ESPE shall remain in normal operation.		P
	During test 3, it is permissible that the OSSDs go to the OFF state when applying the interruption. If a lock-out condition occurs, the ESPE shall be recovered from the lock-out condition before continuing the test.	In the test, monitoring the output distance instead of the status of the OSSD.	P
	Where the same lock-out condition occurs for 3 times, the test is deemed as passed and no further interruptions are required in this state.		P
5.4.3.3	Burst		-
5.4.3.3.1	General tests		-
	For coupling of DC or less than 50 V AC power ports and signal ports, a capacitive coupling clamp according IEC 61000-4-4:2012 shall be used. For other AC power ports, a coupling device network (CDN) according IEC 61000-4-4:2012 shall be used.	DC power input	P
	The ESPE shall be subjected to bursts as specified in 4.3.2.3.1. A B test shall be performed in accordance with following test sequence.	Tested	P
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		P
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	3) The disturbances in accordance with 4.3.2.3.1 shall be applied with a duration of at least 1 min per exposure. During the application, it shall be verified that the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state.		P
	5) The disturbance in accordance with 4.3.2.3.1 shall be applied again with a duration of at least 1 min per exposure. During the exposure, it shall be verified that the OSSD(s) remain(s) in the ON-state.		P
	6) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	The requirements specified in 4.3.2.3.1 consists of several exposures (different coupling methods, voltages, polarities). It is acceptable to run through the specified test sequence only once with all exposures applied sequentially in step 3 and again in step 5, or the test sequence may be run through several times with only one or a selection of exposures applied in each run.		P
5.4.3.3.2	Additional tests		-
	A type 3 or a type 4 ESPE shall also be subjected to burst as specified in 4.3.2.3.2. A C test shall be performed as follows.	Type 1 ESPE	N/A
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		N/A
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		N/A
	3) The disturbance in accordance with 4.3.2.3.2 shall be applied. For a type 3 ESPE, the test duration shall be at least 1 min per exposure. For a type 4 ESPE, the time shall be at least 5 min per exposure. During the application, it shall be verified that no failure to danger occurs. If a lock-out condition occurs, the ESPE shall be recovered from the lock-out condition and the same exposure shall be applied three times (i.e. repeat steps 1 to 3 three times). After this, the test shall be continued with the next exposure.		N/A
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state, or, in case a lock-out condition occurred, the ESPE shall resume normal operation following recovery from a lock-out condition and steps 1 to 4 shall be repeated three times.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The requirements specified in 4.3.2.3.2 consists of several exposures (different coupling methods, voltages, polarities). It is permissible to run through the specified test sequence only once with all exposures applied sequentially in step 3, or the test sequence may be run through several times with only one or a selection of exposures applied in each run.		N/A
	If the ESPE is going to an OFF- state or lock-out condition because of destroyed components, it shall be verified by analysis that no failure to danger condition will occur as a result of destroyed components. In this case, the test is passed.		N/A
5.4.3.4	Surge		-
5.4.3.4.1	General tests		-
	For coupling of DC or less than 50 V AC power ports, a coupling network in accordance with Figure 9 of IEC 61000-4-5:2014 shall be used. For coupling to other AC power ports, a coupling network in accordance with Figure 5 or 7 and Figure 6 or 8 of IEC 61000-4-5:2014 shall be used. For coupling to signal ports, an appropriate coupling network in accordance with Figure 9, 10 or 11 of IEC 61000-4-5:2014 shall be used. In the case of shielded lines with the shield connected to ground on both sides, direct coupling in accordance with Figure 12 of IEC 61000-4-5:2014 shall be applied.	DC power port	P
	The ESPE shall be subjected to surge as specified in 4.3.2.4.1. A B test shall be performed as follows.		P
	1) The OSSD(s) shall be in the ON-state in absence of a test piece in the detection zone.		P
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	3) The disturbance in accordance with 4.3.2.4.1 shall be applied with at least five pulses per exposure. During the application, it shall be verified that the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.		P
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	5) The disturbance in accordance with 4.3.2.4.1 shall be applied again with at least five pulses per exposure. During the application, it shall be verified that the OSSD(s) remain(s) in the ON-state.		P
	6) The test piece shall be placed into the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	The requirements in 4.3.2.4.1 consist of several exposures (different coupling methods, voltages, polarities). It is permissible to run through the specified test sequence only once with all exposures applied sequentially in step 3 and again in step 5, or the test sequence may be run through several times with only one or a selection of exposures applied in one run.		P
5.4.3.4.2	Additional tests		-
	The type 3 or a type 4 ESPE shall also be subjected to surges as specified in 4.3.2.4.2. A C test shall be performed as follows.	Type 1 ESPE	N/A
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		N/A
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		N/A
	3) The disturbance in accordance with 4.3.2.4.2 shall be applied with at least five pulses per exposure. During the application, it shall be verified that no failure to danger occurs. If a lock-out condition occurs, the ESPE shall be recovered from the lock-out condition and the same exposure shall be applied three times (i.e. repeat steps 1 to 3 three times). After this, the test shall be continued with the next exposure.		N/A
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state, or, in case a lock-out condition occurred, the ESPE shall resume normal operation following recovery from a lock-out condition and steps 1 to 4 shall be repeated three times.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The requirements in 4.3.2.4.2 consist of several exposures (different coupling methods, voltages, polarities). It is permissible to run through the specified test sequence only once with all exposures applied sequentially in step 3, or the test sequence may be run through several times with only one or a selection of exposures applied in one run.		N/A
	If the ESPE is going to an OFF-state or lock-out condition because of destroyed components, it shall be verified by analysis that no failure to danger condition will occur as a result of destroyed components. In this case, the test is passed.		N/A
5.4.3.5	Electromagnetic field		-
5.4.3.5.1	General tests		-
	The ESPE shall be subjected to an electromagnetic field as specified in 4.3.2.5.1.		P
	A B test shall be performed as follows.	Tested	P
	1) The OSSD(s) shall be in the ON-state in absence of a test piece in the detection zone.		P
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	3) The disturbance in accordance with 4.3.2.5.1 shall be applied. During the application, it shall be verified that the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.		P
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state.		P
	5) The disturbance in accordance with 4.3.2.5.1 shall be applied again. During the application, it shall be verified that the OSSD(s) remain(s) in the ON-state.		P
	6) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
5.4.3.5.2	Additional tests		-
	A type 3 and type 4 ESPE shall also be subjected to an electromagnetic field as specified in 4.3.2.5.2.	Type 1 ESPE	N/A
	A C test shall be performed as follows.		N/A
	1) The OSSD(s) shall be in the ON-state in absence of a test piece in the detection zone.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	2) The test piece shall be placed into the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		N/A
	3) The disturbance in accordance with 4.3.2.5.2 shall be applied. During the application, it shall be verified that no failure to danger occurs. If a lock-out condition occurs, the ESPE shall be recovered from the lock-out condition and the same exposure shall be continued with the next test frequency.		N/A
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state, or, in case a lock-out condition occurred, the ESPE shall resume normal operation following recovery from a lock-out condition.		N/A
	If the ESPE is going to an OFF-state or lock-out condition because of destroyed components, it shall be verified by analysis that no failure to danger condition will occur as a result of destroyed components. In this case, the test is passed.		N/A
5.4.3.6	Conducted disturbances induced by radio-frequency fields		-
5.4.3.6.1	General tests		-
	The ESPE shall be subjected to conducted radio-frequency disturbances as specified in 4.3.2.6.1.		P
	A B test shall be performed as follows.	Tested	P
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		P
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	3) The disturbance in accordance with 4.3.2.6.1 shall be applied. During the application, it shall be verified that the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.		P
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state.		P
	5) The disturbance in accordance with 4.3.2.6.1 shall be applied again. During the application, it shall be verified that the OSSD(s) remain(s) in the ON-state.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	6) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
5.4.3.6.2	Additional tests		-
	A type 3 or a type 4 ESPE shall also be subjected to conducted radio-frequency disturbances as specified in 4.3.2.6.2. A C test shall be performed as follows.	Type 1 ESPE	N/A
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		N/A
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		N/A
	3) The disturbance in accordance with 4.3.2.6.2 shall be applied. During the application, it shall be verified that no failure to danger occurs. If a lock-out condition occurs, the ESPE shall be recovered from the lock-out condition and the same exposure shall be continued with the next test frequency.		N/A
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state, or, in case a lock-out condition occurred, the ESPE shall resume normal operation following recovery from a lock-out condition.		N/A
	If the ESPE is going to an OFF-state or lock-out condition because of destroyed components, it shall be verified by analysis that no failure to danger condition will occur as a result of destroyed components. In this case, the test is passed.		N/A
5.4.3.7	Electrostatic discharge		-
5.4.3.7.1	General tests		-
	The ESPE shall be subjected to electrostatic discharge as specified in 4.3.2.7.1.		P
	A B test shall be performed as follows.	Tested	P
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		P
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	3) The disturbance in accordance with 4.3.2.7.1 shall be applied with at least 10 pulses per exposure. During the application, it shall be verified that the OSSD(s) remain(s) in the OFF-state with the test piece being present in the detection zone.		P
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state.		P
	5) The disturbance in accordance with 4.3.2.7.1 shall be applied again with at least 10 pulses per exposure. During the application, it shall be verified that the OSSD(s) remain(s) in the ON-state.		P
	6) The test piece shall be inserted into the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		P
	The requirements in 4.3.2.7.1 consist of several exposures (e.g. different locations). It is permissible to run through the specified test sequence only once with all exposures applied sequentially in step 3 and again in step 5, or the test sequence may be run through several times with only one or a selection of exposures applied in one run.		P
5.4.3.7.2	Additional tests		-
	A type 3 or a type 4 ESPE shall also be subjected to electrostatic discharge as specified in 4.3.2.7.2. A C test shall be performed as follows.	Type 1 ESPE	N/A
	1) The OSSD(s) shall be in the ON-state in the absence of a test piece in the detection zone.		N/A
	2) The test piece shall be placed in the detection zone. The OSSD(s) shall respond by going from the ON-state to the OFF-state.		N/A
	3) The disturbance in accordance with 4.3.2.7.2 shall be applied. For a type 3 ESPE at least 10 pulses per exposure shall be applied. For a type 4 ESPE at least 30 pulses per exposure shall be applied. During the application, it shall be verified that no failure to danger occurs. If a lock-out condition occurs, the ESPE shall be recovered from the lockout condition and the same exposure shall be applied three times (i.e. repeat steps 1 to 3 three times). After this, the test shall be continued with the next exposure.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	4) The test piece shall be removed from the detection zone. The OSSD(s) shall respond by going from the OFF-state to the ON-state, or, in case a lock-out condition occurred, the ESPE shall resume normal operation following recovery from a lock-out condition and steps 1 to 4 shall be repeated three times.		N/A
	The requirements in 4.3.2.7.2 consist of several exposures (different locations). It is permissible to run through the specified test sequence only once with all exposures applied sequentially in step 3, or the test sequence may be run through several times with only one or a selection of exposures applied in one run.		N/A
	If the ESPE is going to an OFF-state or lock-out condition because of destroyed components, it shall be verified by analysis that no failure to danger condition will occur as a result of destroyed components. In this case, the test is passed.		N/A
5.4.4	Mechanical influences		-
5.4.4.1	Vibration		-
	The test sample shall be exposed to vibration tests in accordance with IEC 60068-2-6 (Fc: vibration sinusoidal). If information for use specifies that anti-vibration mounts are required, then they shall be used during the tests.	Tested	P
	For stationary use, conditions in Table 5 apply.	Class 3M4	P
	If available classes in accordance with Table 5 are not representative for the application, the tests shall be carried out in accordance with the supplier's specified alternative conditions.		N/A
	The following limited functional tests shall be performed for each axis:		P
	- an A test during each of the first and last cycles;		P
	- a B test, so that the test piece is inserted into the detection zone at the beginning of the second cycle and is removed at the end of the ninth cycle.		P
	For ground vehicle installations, the conditions given in Table 6 apply.	Class 5M1	P
	If available classes in accordance with Table 6 are not representative for the application, the tests shall be carried out in accordance with the supplier's specified alternative conditions.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The following limited functional tests shall be performed for each axis:		P
	- an A test during each of the first and last cycles;		P
	- a B test, so that the test piece is inserted into the detection zone at the beginning of the second cycle and is removed at the end of the ninth cycle.		P
	The test sample shall be exposed to vibration tests in accordance with IEC 60068-2-64 (Fh: vibration broadband random).		P
	For ground vehicle installations, the conditions given in Table 7 apply.	Class 5M1	P
	If available classes in accordance with Table 7 are not representative for the application, the tests shall be carried out in accordance with the supplier's specified alternative conditions.		P
	The following limited functional tests shall be performed for each axis (i.e. three times):		P
	- a B test, so that the test piece is brought into the detection zone before the vibration starts and is removed at the end vibration;		P
	- an A test, after removal of the test piece.		P
5.4.4.2	Shock		-
	The test sample shall be exposed to shock tests in accordance with IEC 60068-2-27 (Ea: shock half-sine). If information for use specifies that anti-vibration mounts are required, then they shall be used during the tests.		P
	For stationary use, the conditions given in Table 8 apply.	Tested	P
	If available classes in accordance to Table 8 are not representative for the application, the tests shall be carried out in accordance with the supplier's specified alternative conditions.		P
	The following tests shall be performed for each axis:		P
	- an A test during each of the first and the last (10 ± 1) shocks;		P
	- a B test, so that the test piece is inserted into the detection zone after the first (10 ± 1) shocks and is removed before the last (10 ± 1) shocks.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The test sample shall be exposed to shock tests in accordance with IEC 60068-2-27 (Ea: shock half-sine).		P
	For ground vehicle installations, the conditions given in Table 9 apply.	Class 5M1 tested	P
	If available classes in accordance to Table 9 are not representative for the application, the tests shall be carried out in accordance with the supplier's specified alternative conditions.		P
	The following tests shall be performed for each axis:		P
	- an A test during each of the first shocks;		P
	- a B test, so that the test piece is brought into the detection zone after the first shock.		P
5.4.5	Enclosures		-
	The requirements of 4.3.4 for degrees of protection shall be tested in accordance with IEC 60529. The remaining requirements shall be verified by inspection.	IP67 reached	P
5.4.6	Light interference		-
5.4.6.1	General		-
	Each test shall be carried out at an operating distance of 3 m (or the closest normal operating distance to 3 m as specified by the supplier) and under the stated conditions as a minimum requirement. Additional tests shall be carried out under different combinations of operating distances and environmental conditions when		P
	- the supplier states higher immunity levels, which shall be verified by testing at those levels with appropriate light sources, and/or	No this situation	N/A
	- an analysis shows that additional tests are necessary.	No this situation	N/A
	The particular test procedures shall be as defined in a subsequent part of IEC 61496 based on the sensing technology used.	IEC 61496-3	P
5.4.6.2	Light sources		-
	The light sources shall be as follows.		-
	a) Incandescent light source: a linear tungsten halogen (quartz) lamp with the following characteristics:		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- colour temperature: 2 800 K to 3 200 K;		P
	- rated voltage: any value within the range 100 V to 250 V;		P
	- supply voltage: rated voltage ±5 %, sinusoidal AC at 48 Hz to 62 Hz.		P
	The lamp can be mounted in a parabolic reflector of minimum dimensions 200 mm × 150 mm, having a diffuse reflective surface and a reflectivity that is uniform within ±5 % over the wavelength range 400 nm to 1 500 nm.		P
	NOTE This source produces a beam of near-uniform intensity with known spectral distribution and having a predictable modulation at twice the supply frequency.		-
	b) Fluorescent light source: a linear fluorescent tube with the following characteristics:		P
	- size: T8 × 600 mm minimum;		P
	- colour temperature: 3 000 K to 6 500 K; used in combination with an electronic ballast having the following characteristics:		P
	- operating frequency: 25 kHz to 50 kHz;		P
	- power rating corresponding to the tube; and operated at its rated power supply voltage ±5 %, without a reflector or diffuser.		P
	Other fluorescent light sources having, for example, electronic ballasts with an operating frequency other than that specified can lead to different test results. Therefore, the use of other types of fluorescent light sources or a light source generator simulating the effects of different fluorescent light sources should be considered for testing.		P
	c) Flashing beacon light source: a light source employing a xenon flash tube (without enclosure, reflector or filter) having the following characteristics:		P
	- flash duration: from 40 µs to 1 200 µs (measured to the half-intensity point);		P
	- flash frequency: 0,5 Hz to 2 Hz;		P
	- input energy per flash: 3 J to 5 J;		P
	d) Laser beam pointer: a collimated laser beam having characteristics within the following limits:		P
	- flash duration: continuous wave mode;		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- wavelength: within 550 nm up to 670 nm;		P
	- beam shape: diameter below 5 mm;		P
	- light intensity 0,7 mW up to 1 mW;		P
	- laser class 2.		P
	WARNING - The laser beam pointer is a class 2 device. Never look into the beam.		-
	The equipment setups and test procedures shall be as defined in a subsequent part of IEC 61496 based on the sensing technology being used. For some sensing technologies, some of these light sources can be not relevant. For some technologies, tests with other light sources can be required	IEC 61496-3	P
5.4.6.3	Test sequences		-
	Test sequence 1. a) OSSDs of the ESPE in ON-state. b) Switch on interfering light (the OSSDs shall remain in ON-state). c) B test. d) Switch off ESPE for 5 s. Restore power. Reset start interlock if fitted. e) B test. f) Switch off interfering light. g) B test.		P
	Test sequence 2. a) OSSDs of the ESPE in ON-state. b) Switch on interfering light. c) C tests repeatedly for 1 min. d) Switch off the ESPE for 5 s. Restore power. Reset start interlock if fitted. e) C tests repeatedly for 1 min. f) Switch off interfering light. g) C tests repeatedly for 1 min.		P
	Test sequence 3. a) OSSDs of the ESPE in ON-state. b) Switch on interfering light. c) C tests repeatedly for 3 min.		P
	If a system analysis proves that a switch-off of the ESPE and a system reset lead to identical system behaviour, then the switch-off of the ESPE may be replaced by a system reset.		P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
5.5	Validation of programmable or complex integrated circuits		-
5.5.1	General		-
	This subclause 5.5 deals with the validation of the requirements of 4.2.10 and 4.2.11, and any analysis statement included as part of the test results statement required by 5.3.1.		N/A
	Validation shall be undertaken by a competent person(s) who should be independent of those responsible for any aspect of the system design, the hardware design and the software design. A written validation report shall be compiled.		N/A
	NOTE The validation provides independent confirmation that specific requirements have been achieved. The process is intended to confirm that systematic faults in the design have been avoided, that procedures are in place to maintain safety performance during the life cycle of the product (including, for example, following modification) and that the design of the ESPE fulfils the fault detection requirements appropriate to its type.		-
5.5.2	Complex or programmable integrated circuits		-
	For a Type 4 ESPE employing a complex or programmable integrated circuit(s), the following requirements shall be validated by analysis:	Type 1 ESPE	N/A
	a) there are a minimum of two independent decision-making channels;		N/A
	b) the detection of disparity between channels, and the initiation of a lock-out condition, shall be maintained under all applicable fault conditions.		N/A
5.5.3	Software, programming, functional design of integrated circuits		-
	Verification and validation shall be in accordance with the standard(s) selected for development (see 4.2.11.2).		N/A
5.5.4	Test results analysis statement		-
	When analysis is employed to define the result of any test required by 5.3, the adequacy, suitability, and validity of the techniques used shall be validated. The correct implementation of the methods used shall be verified by repeating parts of the analysis selected at random.		N/A
6	Marking for identification and for safe use		-
6.1	General		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	In accordance with ISO 12100, all parts of the ESPE shall bear all markings which are necessary:	See nameplate	P
	- for its unambiguous identification;		P
	- for its safe use,		P
	and supplementary information shall be given, as appropriate:		P
	- permanently on the ESPE;		P
	- in accompanying documents such as instruction handbooks;		P
	- on the packaging.		P
	The enclosure of the most appropriate part of the ESPE shall carry the following permanent markings:		P
	a) identification of the product, including name and address of the supplier, designation of series or type, serial number and year of construction;		P
	b) parameters, for example dimensions, of the detection zone;		P
	c) detection capability;		P
	d) response time;		P
	e) rated voltage(s) including number of phases and frequency where relevant;		P
	f) rated input power (if greater than 25 W) or rated current;		P
	g) designation of IP code;		P
	h) for class II and class III equipment only, symbol for classification for protection against electric shock (for class II the symbol is IEC 60417-5172:2003-02 and for class III the symbol is IEC 60417-5180:2003-02);	Class I	N/A
	i) warning sign of hazards arising from dangerous voltages;	9-36V	N/A
	j) type of ESPE in accordance with 4.1.3;	Type 1 ESPE	P
	k) PL and/or SIL in accordance with 4.1.4.	Not safety ESPE	N/A
6.2	ESPE supplied from a dedicated power supply		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	Where an ESPE is designed to be supplied from an external dedicated power supply, details of the model or type of dedicated power supply with which it has been tested shall be permanently marked on the enclosure of the most appropriate part of the ESPE and/or included in the instructions for use.	9V-36V DC	P
6.3	ESPE supplied from an internal electrical power source		-
	An ESPE supplied from an internal power source shall be marked with details of the rated current of the supply fuse, if applicable, on the enclosure of the most appropriate part of the ESPE.	No this situation	N/A
6.4	Adjustment		-
	When the ESPE can be adjusted to suit different rated voltages or different inputs, a marking showing the voltage or input to which the ESPE is adjusted shall be clearly and easily discernible at the point of adjustment.	9V-36V DC	P
6.5	Enclosures		-
	Any enclosure which contains electrical devices that can give rise to a risk of electric shock shall be marked with a warning sign in accordance with 16.2.1 of IEC 60204-1:2016.	No this situation	N/A
6.6	Control devices		-
6.6.1	Markings for switches, indicating lamps, and other control devices shall be placed adjacent to those components; they shall not be placed on removable parts which can be replaced in such a way that the marking is misleading.		P
6.6.2	Functional identification of control and indicating devices shall be in accordance with 16.3 of IEC 60204-1:2016.		P
6.6.3	The positions of any power supply switches shall be marked in accordance with 5.3.1 of IEC 60204-1:2016.		N/A
6.6.4	Devices intended for the adjustment of a characteristic during or after installation, shall be provided with markings of the direction of adjustment to increase or decrease the value of that characteristic. See also IEC 60447.	No this device	N/A
6.7	Terminal markings		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
6.7.1	Terminals to which cable connections are to be made at the time of installation, or re-established after maintenance of the ESPE, shall be marked and related to a diagram.	No this device	N/A
6.7.2	Terminals for external connections provided with, and relating to the ESPE and user replaceable components shall be marked and related to a diagram.	No this device	N/A
6.7.3	All terminals for the incoming supply connections shall be marked in accordance with IEC 60445.	No this device	N/A
6.7.4	Protective conductor connection points shall be marked in accordance with 8.2.4 of IEC 60204-1:2016.	No this device	N/A
	This marking shall not be placed on screws, removable washers or other parts which might be removed when conductors are being connected/disconnected.	No this device	N/A
6.7.5	Where an ESPE is to be connected to more than two supply conductors, it shall be provided with a connection diagram, fixed to the ESPE, unless the correct mode of connection is obvious.	No this situation	N/A
6.7.6	If more than one supply is taken to an ESPE, the marking shall include a warning that all supplies shall be switched off before the terminal cover may be removed.	No this situation	N/A
6.8	Marking durability		-
	Marking shall be capable of withstanding the environmental influences of an industrial environment with respect to temperature and humidity as defined by this document, and liquids such as water, soapy water, machine oil, benzine, etc.		P
	Markings shall be capable of withstanding being rubbed lightly for 15 s with a piece of cloth soaked with petroleum spirits, and with a piece of cloth soaked in water.		P
7	Accompanying documents		-
	The supplier of the ESPE shall provide documentation in the languages agreed between the user and the supplier.	English version	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The accompanying documents shall contain the information required for the installation, use and subsequent disposal of the ESPE, including the following information where applicable:	See the manual	P
	a) a statement that other devices shall not be connected to the internally generated power supplies of the ESPE;	No internally generated power supplies	N/A
	b) details of the included optional functions of the ESPE, as described in Annex A (including all necessary data for determining safety performance);	No these functions	N/A
	c) a description of facilities for connection of Stopping Performance Monitor (SPM) if provided;	No this device	N/A
	d) for a type 2 ESPE, information on the means of applying an external test signal, where required (see 4.2.2.3);	Type 1 ESPE	N/A
	e) a recommendation that security keys/special tools, if provided for adjustments, operations or access, be kept under the control of a responsible or authorized person(s);	Not safety type	N/A
	f) the size and type of the test piece and the test procedure, or description of other methods for checking the detection capability and the operation of the visual indicator;	See the manual	P
	g) the response time. If a safety-related communication interface is used, procedures for determining overall system response time;	See the manual	P
	h) the rated operating conditions for the ESPE, including - temperature range, - humidity, - voltage range, and - range of separation distances between the subsystems, and the maximum length of interconnecting cables;	See the manual	P
	i) advice on the prevention of mutual interference of sensing functions;	See the manual	P
	j) block diagrams, functional description chart showing sequence of relay switching operations;	See the manual	P
	k) the location of all input and output terminals;	See the manual	P
	l) the ratings and characteristics of all input/output terminals;	See the manual	P

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	m) the minimum and maximum voltage and current that the OSSDs (and if provided, the SSDs) are capable of switching into a resistive, capacitive or inductive load, and the maximum switching rate with this load, and the anticipated life of the switching device depending on load;	No OSSDs	N/A
	n) information to enable the user to carry out maintenance using the supplier's recommended spares;	See the manual	P
	o) special requirements regarding input cables and terminations, if applicable;	See the manual	P
	p) the total load/power requirements of the ESPE;	See the manual	P
	q) details of the space required around the equipment for removal and maintenance purposes;	See the manual	P
	r) a list of the user-replaceable parts specified by the supplier;	See the manual	P
	s) a list of colour and coding systems (see IEC 60204-1:2016);	See the manual	P
	t) the overall dimensions of the equipment;	See the manual	P
	u) operating instructions;	See the manual	P
	v) the location and dimensions of the detection zone(s) and definitions of other functional limits;	See the manual	P
	w) a schedule of checks to be carried out after installation, after maintenance, or periodically, to establish that the device is functioning correctly;	See the manual	P
	x) the method and frequency of regular testing to confirm that proper operation is maintained;	See the manual	P
	y) a statement of the IP rating of the enclosures, or where the ESPE is intended to be mounted in a machine controlgear enclosure, the minimum IP rating required of that enclosure in accordance with 4.3.4;	See the manual	P
	z) a clear statement of any particular application or limited intended use for which the ESPE is intended;	See the manual	P
	aa) for a type 2 ESPE when the periodic test is externally initiated, information on selecting the required test interval; if the periodic test is internally initiated, information on the internal test interval;	Type 1 ESPE	N/A
	bb) the installation and mounting instructions for any switches, controls and indicators remotely mounted from the ESPE that are connected to the ESPE;	No remote devices	N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	cc) instructions on where parts with restart interlock should be located in relation to the danger zone;		N/A
	dd) instructions on where parts with sensing functions should be located in relation to the danger zone and methods for determination of safety distances between those parts and the danger zones, for example formulae for calculation;	See the manual	P
	ee) instructions on how the ESPE should be interfaced with the machine control system (e.g. time OSSD will remain in the OFF-state after the sensor is de-activated, the characteristics of test pulse timing);	No OSSDs	N/A
	ff) the details of any special precautions necessary to be taken into account;	See the manual	P
	gg) the dimensions of the space(s) to be provided for the ESPE;	See the manual	P
	hh) the dimensions and position of the means for supporting and fixing the ESPE within the space(s);	See the manual	P
	ii) the minimum clearances between the various parts of the ESPE and the surrounding parts of the fitment;	See the manual	P
	jj) the means of connection of the ESPE to the supply, and the interconnection of separate components, if any;	See the manual	P
	kk) information for proper connection of solid state outputs in accordance with 4.2.4.3, if used. When a communications interface (4.2.4.4) is integrated into the ESPE, the operating limitations and timing characteristics necessary for proper integration shall be provided;	No OSSDs	N/A
	ll) when blanking functions (monitored, unmonitored, fixed or floating) or adjustment of detection capability are provided, their intended use shall be explained;	No this function	N/A
	mm) type of ESPE, PL and/or SIL and the PFHD (see IEC 62061 and/or ISO 13849-1) or other relevant information such as reliability data shall be provided together with necessary information to take into account components that have life affected by application (e.g. maximum number of operations and load characteristics).	See the manual	P
Annex A	Optional functions of the ESPE		-
A.1	General		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	An ESPE may include additional functions, or devices arranged to perform functions within the safety-related control system. The ESPE together with any separate devices specifically intended to perform these optional functions shall meet the requirements of this document.	No additional functions	N/A
	NOTE Where the optional functions are performed by a separate device(s) that is not specifically intended to be used as part of an ESPE, this separate device(s) should meet the relevant requirements of other appropriate standards (for example ISO 13849-1, IEC 61508 (all parts), IEC 62061). In this case, the requirements of this Annex A can be used as a guide in conjunction with the other standards.		-
	The following are optional devices or functions: - external device monitoring (see Clause A.2); - stopping performance monitor (see Clause A.3); - secondary switching device (for type 2 ESPEs) (see Clause A.4); - start interlock (see Clause A.5); - restart interlock (see Clause A.6); - muting (see Clause A.7); - ESPE used as a machine reinitiation device (see Clause A.8).	No this functions	N/A
	The following requirements are minimum requirements and may not be sufficient for all applications. These requirements are intended to be used in conjunction with the relevant requirements given in other standards (for example IEC 60204-1, ISO 13849-1), as the result of a risk assessment.		N/A
	Where signals for optional functions are provided via a safety-related data interface, the requirement for hard-wired connections at the ESPE can be excluded if the equivalent functions are performed by a safety-related communication system (see also 4.2.4.4).		N/A
A.2	External device monitoring (EDM)		N/A
A.2.1	Functional requirements		-
	EDM shall provide a means to monitor the state of external contacts (for example FSDs or MPCEs).		N/A
	The ESPE shall go to a lock-out condition when an incorrect state is detected in one of the devices being monitored by the EDM.		N/A
A.2.2	Fault condition requirements		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The ESPE shall respond in accordance with 4.2.2 to faults in the EDM.		N/A
A.2.3	Verification		-
	It shall be verified by inspection and test that:		N/A
	- the necessary means are provided in the ESPE for the monitoring of the devices for which the EDM is intended to be used;		N/A
	- the ESPE goes to the lock-out condition when any one of the devices being monitored is in an incorrect state;		N/A
	- the ESPE responds in accordance with 4.2.2 to faults occurring in the EDM.		N/A
A.2.4	Information for use		-
	The ESPE supplier shall provide information for the connection of the EDM to the appropriate devices. The supplier shall specify the type of any device for which the EDM is intended to be used. Where the monitored devices require particular characteristics (for example mechanically linked contacts, duplicate inputs, N/O, N/C), these shall be specified.		N/A
	Unless response time of the external contacts is monitored, information for use shall include a notice that external means of monitoring the response time of the contacts can be required.		N/A
A.3	Stopping performance monitor (SPM)		N/A
A.3.1	Functional requirements		-
A.3.1	Functional requirements		-
	The SPM shall provide signals to the ESPE related to the time taken by, or the amount of travel of, the hazardous parts of the machine in coming to rest or reverting to a safe condition. The ESPE shall go to a lock-out condition when a signal(s) from the SPM indicates that the preset limit of stopping performance has been exceeded.		N/A
	For a type 3 or a type 4 ESPE, the SPM shall provide at least two signalling channels to the ESPE. Each channel shall be capable of initiating a lock-out condition of the ESPE.		N/A
	The SPM shall apply an automatic stopping performance test to monitor the overall system stopping performance.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The SPM shall be capable of initiating the automatic stopping performance test in response to signals derived from the ESPE, immediately upon actual or simulated actuation of the sensing device.		N/A
	Any means by which the preset limit(s) within the SPM may be adjusted shall require the use of a key, keyword or special tool.		N/A
A.3.2	Fault condition requirements		-
	The ESPE shall go to a lock-out condition in response to any of the following:		N/A
	- upon failure to apply, or to complete the automatic test;		N/A
	- upon failure of transmission of motion to the SPM or, when duplicated transmission means are used, upon failure of either one of those means;		N/A
	- upon disconnection of the stopping performance monitor from the ESPE.		N/A
A.3.3	Verification		-
	Verify by inspection that:		-
	- the SPM output signal(s) causes the ESPE to go to a lock-out condition when the preset limit of stopping performance is exceeded;		N/A
	- for a type 3 or a type 4 ESPE application, there are at least two independent signal sources, from the SPM to the ESPE, and failure of either one of those causes lock-out;		N/A
	- the SPM initiates an automatic test in response to a signal from the ESPE;		N/A
	- the ESPE initiates an automatic stopping performance test upon actual or simulated actuation of the sensing function;		N/A
	- any adjustment means requires the use of a key, keyword or special tool;		N/A
	- when the automatic test has failed to be applied or completed, a lock-out condition is achieved;		N/A
	- when any of the transmission of motion means has failed, a lock-out condition is achieved;		N/A
	- when a SPM is disconnected from the ESPE, or from the safety-related control system, a lock-out condition is achieved;		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- the markings conform to A.3.4 and are correct.		N/A
A.3.4	Marking		-
	The supplier shall provide permanently affixed markings to the stopping performance monitor giving the following information:		N/A
	- name and address of the supplier;		N/A
	- the model type number and serial number;		N/A
	- the ESPE type number for which the stopping performance monitor is designed;		N/A
	- the accuracy of the unit.		N/A
A.4	Secondary switching device (SSD)		-
A.4.1	Functional requirements		-
	When the ESPE power supply is switched off or when the ESPE is in a lock-out condition, the SSD shall be in the OFF-state.		N/A
	The ability of the SSD to perform its safety-related function shall be checked by an automatic test, carried out when the ESPE power supply is switched on and before the OSSD(s) go to the ON-state.		N/A
A.4.2	Fault condition requirements		-
	If the automatic test referred to in A.4.1 identifies the inability of the SSD to go to the OFF‑state, the OSSD(s) shall remain in the OFF-state.		N/A
A.4.3	Verification		-
	Verify by inspection and test that:		-
	- when, under fault conditions, the SSD is restrained to remain in the ON-state and the ESPE power supply is switched on, the OSSD(s) will remain in the OFF-state, even when a reset is attempted;		N/A
	- the SSD is in the OFF-state under lock-out conditions.		N/A
A.5	Start interlock		-
A.5.1	Functional requirements		-
	The start interlock shall prevent the OSSD(s) going to the ON-state when the electrical supply is switched on, or is interrupted and restored.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The OFF-state of the OSSD(s) shall continue until the start interlock is reset to its ON-state, manually (for example by switch operation or by actuation and de-actuation of the sensing device).		N/A
	It shall not be possible for reset of a start interlock to reset the OSSD(s) to the ON-state under a lock-out condition.		N/A
A.5.2	Fault condition requirements		-
	A failure of the start interlock which causes it to go to, or remain in a permanent ON-state shall cause the ESPE to go to, or to remain in the lock-out condition.		N/A
A.5.3	Verification		-
	Verify by inspection and test that:		-
	- the OSSD(s) is in the OFF-state when the start interlock is in the OFF-state;		
	- when the power supply is switched on, the OSSD(s) remain in the OFF-state until the start interlock is manually operated;		N/A
	- when power is interrupted long enough for the OSSD(s) to go to the OFF-state, it shall be verified that when the power is restored, the OSSD(s) remain in the OFF-state until the start interlock is manually operated;		N/A
	- during a lock-out condition, an attempt to reset the start interlock does not allow the OSSD(s) to go to the ON-state;		N/A
	- the lock-out condition is initiated if the start interlock fails.		N/A
A.5.4	Indication		-
	A yellow indicator shall be provided which should illuminate when the OSSD(s) is prevented from going to the ON-state by the start interlock.		N/A
A.6	Restart interlock		-
A.6.1	Functional requirements		-
	The restart interlock shall prevent the OSSD(s) going to the ON-state when:		N/A
	- the detection zone is interrupted while the machine operation is at a hazardous part of its operating cycle;		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- the detection zone is interrupted while the machine is in automatic or semi-automatic mode;		N/A
	- there is a change of the machine operating mode or type of operation.		N/A
	The interlock condition shall continue until the restart interlock is manually reset. However, it shall not be possible to reset the restart interlock whilst the sensing device is actuated.		N/A
A.6.2	Fault condition requirements		-
	A failure to meet the functional requirements of A.6.1 shall cause the ESPE to go to the lockout condition.		N/A
A.6.3	Verification		-
	Verify by inspection and test that:		-
	- the OSSD(s) is in the OFF-state when the restart interlock is in the OFF-state;		N/A
	- the restart interlock will not reset to the ON-state while the sensing device is actuated;		N/A
	- the restart interlock goes to the OFF-state on sensing device actuation during a hazardous machine operation;		N/A
	- when the machine operating mode or type of operation is changed, the restart interlock goes to the OFF-state;		N/A
	- the lock-out condition is initiated if the restart interlock fails.		N/A
A.6.4	Indication		-
	A yellow indicator shall be provided which should illuminate when the OSSD(s) is prevented from going to the ON-state by the restart interlock.		N/A
A.7	Muting		-
A.7.1	General		-
	Subclauses A.7.2 to A.7.5 are applicable only when the methodology of using at least two hard-wired signals to automatically initiate and terminate the mute function is available. Other methods are possible and this Clause A.7 can be used as a guide.		N/A
A.7.2	Functional requirements		-

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
A.7.2.1	When the ESPE is in a muted condition, the OSSD(s) shall remain in the ON-state on actuation of the sensing device.		N/A
A.7.2.2	There shall be at least two independent hard-wired muting signal sources to initiate the function. It shall not be possible to initiate muting when the OSSDs are already in the OFF-state.		N/A
A.7.2.3	The mute function shall only be initiated by the correct sequence and/or timing of the mute signals. Should conflicting muting signals occur, the ESPE shall not allow a muted condition to occur.		N/A
A.7.2.4	There shall be at least two independent hard-wired muting signal sources to stop the function. The muting function shall stop when the first of these muting signals changes state. The deactivation of the muting function shall not rely only on the clearance of the ESPE.		N/A
	NOTE The signal sources for initiating and stopping the mute function can be the same.		-
A.7.2.5	An incorrect sequence and/or the expiration of a pre-set time limit shall cause the termination of the mute condition.		N/A
	NOTE For some applications (for example, conveyor and packaging machinery) a mute dependent manual override function can be provided. Details about the manual override function are provided in IEC 62046.		-
A.7.3	Fault condition requirements		-
	A fault in the muting function shall be detected in accordance with 4.2.2, and shall at least not allow another muted condition to occur. Necessary fault detection of the muting function shall be carried out automatically.		N/A
A.7.4	Verification		-
	Verify by inspection and test that:		-
	- in a muted condition, the OSSD(s) remain in the ON-state on actuation of the sensing device;		N/A
	- there are two independent hard-wired muting signal sources to initiate and to stop the muting function and that a muted condition is prevented whenever an invalid combination of signals is present;		N/A
	- in a type 2 ESPE, any failure which could lead to a continuously muted condition is revealed by the periodic test, and when such a failure is detected, a further muted condition is prevented;		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- the muting function terminates when the first of the "stop muting signal sources" changes state.		N/A
A.7.5	Indication		-
	A mute status signal or indicator shall be provided (in some applications, an indication signal of muting is necessary (see ISO 13849-1)).		N/A
A.8	Reinitiation of machine operation facility		-
A.8.1	General		-
	If, in addition to its function as a protective device, the ESPE is intended to be used to reinitiate machine operation, the following modes of operation can be used:		N/A
	- an actuation and de-actuation of the sensing device reinitiates machine movement, referred to as single break;		N/A
	- two consecutive actuations and de-actuations of the sensing device reinitiate machine movement, referred to as double break.		N/A
	Where this optional feature is provided as part of the ESPE, a start interlock as specified in Clause A.5, and a restart interlock as specified in Clause A.6, shall also be provided within the ESPE.		N/A
A.8.2	Functional requirements		-
	a) When the ESPE power is switched on or is interrupted and restored, it shall not be possible to use either of the modes of operation described in A.8.1 before the start interlock has been reset.		N/A
	b) After actuation of the sensing device during hazardous motion, it shall not be possible to use either of the modes of operation described in A.8.1 before the restart interlock has been reset.		N/A
	c) Successive reinitiations of machine operation by the use of either of the modes of operation described in A.8.1 shall be possible only within a limited period of time.		N/A
	d) When double break reinitiation is selected, reinitiation by single break shall not be possible under any sequence of events or actions.		N/A
	e) If the limited period of time referred to in c) is exceeded, further machine reinitiation shall not be possible until the restart interlock has been reset.		N/A

EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	f) Machine reinitiation by either of the modes of operation described in A.8.1 shall not be possible following a change of reinitiation mode until the restart interlock has been reset.		N/A
	g) Facilities shall be provided to reset the timer controlling the limited period described in c) by external means.		N/A
	h) The adjustment means for the timer shall require the use of a key, keyword or special tool.		N/A
	NOTE The time period permitted for successive reinitiations should not exceed 30 s for machinery having a cycle time of less than 5 s.		-
A.8.3	Fault condition requirements		-
	Any fault listed in Annex B leading to a change in the machine reinitiation mode shall at least result in the operation of either the start interlock or the restart interlock.		N/A
A.8.4	Verification		-
	Verify by inspection and test that:		-
	- after the ESPE power is switched on, or is interrupted and restored, it is not possible to use either of the modes of operation described in A.8.1 until the start interlock has been reset;		N/A
	- after actuation of the sensing device during hazardous motion, it is not possible to use either of the modes of operation described in A.8.1 until the restart interlock has been reset;		N/A
	- successive reinitiations of machine operation by the use of either of the modes of operation described in A.8.1 shall only be possible within a limited time period;		N/A
	- when double break reinitiation is selected, reinitiation by a single break shall not be possible under any sequence of events or actions;		N/A
	- machine reinitiation by either of the modes of operation described in A.8.1 shall not be possible following a change of reinitiation mode until the restart interlock has been reset;		N/A
	- facilities are provided to reset the timer controlling the limited period described above, by external means;		N/A
	- the adjustment means for the timer are within an enclosure with access requiring the use of tools;		N/A

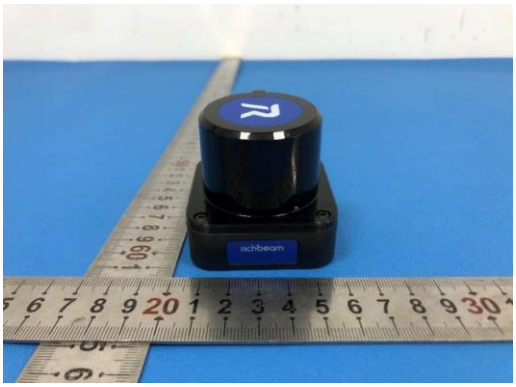
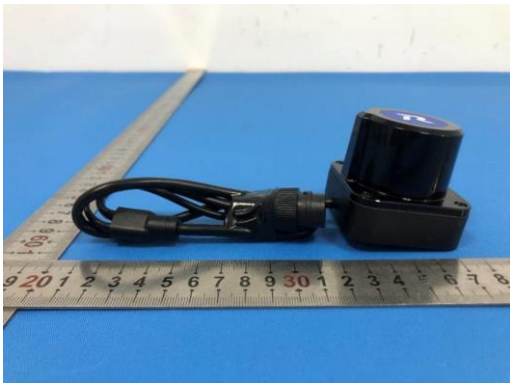
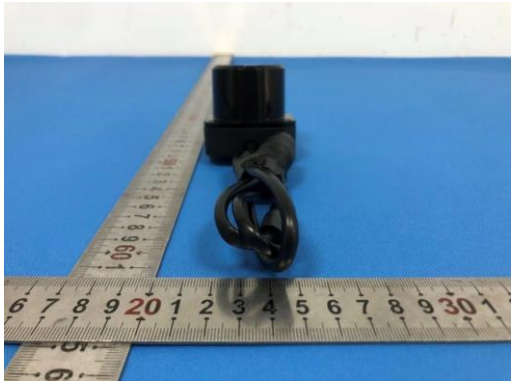
EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	- faults listed in Annex B leading to a change in the machine reinitiation mode shall at least result in operation of either the start interlock or the restart interlock.		N/A
A.9	Setting the detection zone and/or other safety-related parameters		N/A
A.9.1	Functional requirements		-
	The setting of the detection zone and/or other safety-related parameters shall not be possible without using a key, keyword or tool.		N/A
	NOTE For example, the tool can be a password protected software configuration program that is a part of the ESPE.		-
	If the setting is carried out using a personal computer or equivalent fitted with untested dedicated hardware and/or software, a special procedure shall be used for setting the detection zone. This procedure shall be in accordance with appropriate standards (see also 4.2.11). If the tool is software, only software authorized by the supplier shall be used.		N/A
	The setting of safety-related parameters should only be performed by qualified persons. The procedure shall include measures to ensure that the input parameters are transmitted correctly and without corruption to the ESPE. This shall be applied for all safety-related settings, for example, the setting of the response time. The parameterization procedures shall conform to an appropriate standard (e.g. IEC 62061:2005, 6.11.2, or ISO 13849-1:2015, 4.6.4).		N/A
A.9.2	Verification		-
	The setting of a detection zone or other safety-relevant parameter(s) shall be verified as follows:		-
	a) verification of the correct setting function(s) for each configuration parameter (minimum, maximum and representative values) - the possibility of differences between the detection zone as displayed on the screen of a configuration tool (for example, a personal computer) and the actual detection zone of the ESPE should be taken into account;		N/A
	b) verification that the configuration parameters are checked for plausibility, for example by use of invalid values, etc.;		N/A
	c) verification that the access to, and methods of, configuration by the user are in accordance with the requirements of corresponding standards;		N/A

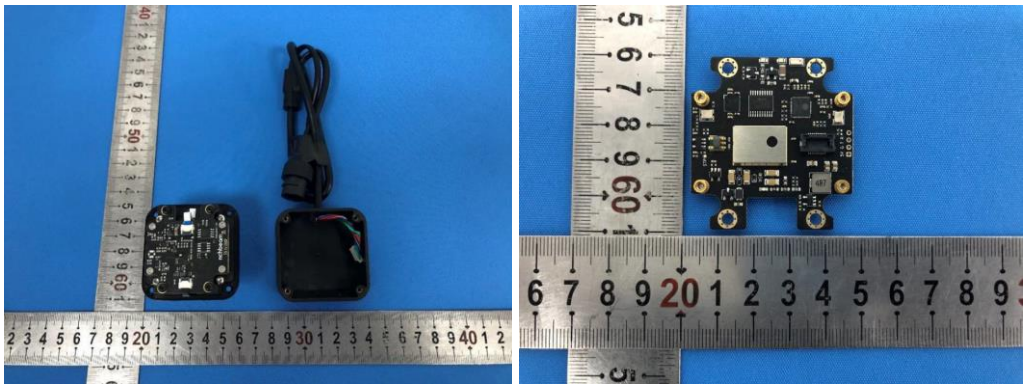
EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	d) verification, in the case of detection zones that can be varied in size during operation, that the data/signals for determining the size of a detection zone are generated and processed in such a way that a single fault shall not lead to a loss of the safety function; verification that such a single fault is detected and causes the OSSDs to remain in the OFF-state or to go to the OFF-state within the response time of the AOPDDR.		N/A
Annex B	Catalogue of single faults affecting the electrical equipment of the ESPE, to be applied as specified in 5.3	Not safety type ESPE	N/A
B.1	General		-
	The faults listed in this Annex B are not exclusive and, if necessary, additional faults shall be considered. For new components not mentioned in Annex B, a failure mode and effects analysis shall be carried out to establish the faults that are to be considered for those components.		N/A
B.2	Conductors and connectors		-
	The requirements of ISO 13849-2:2012, Tables D.4 to D.7, apply.		N/A
B.3	Switches		-
	The requirements of ISO 13849-2:2012, Tables D.8 to D.11, apply.		N/A
B.4	Discrete electrical components		-
	The requirements of ISO 13849-2:2012, Tables D.12 to D.17, apply.		N/A
B.5	Solid-state electrical components		-
	The requirements of ISO 13849-2:2012, Tables D.18 to D.21, apply.		N/A
B.6	Motors		-
	Faults considered		N/A
	Motor stopped		N/A
	Speed above normal		N/A
	Speed below normal		N/A
Annex C	Design review		-

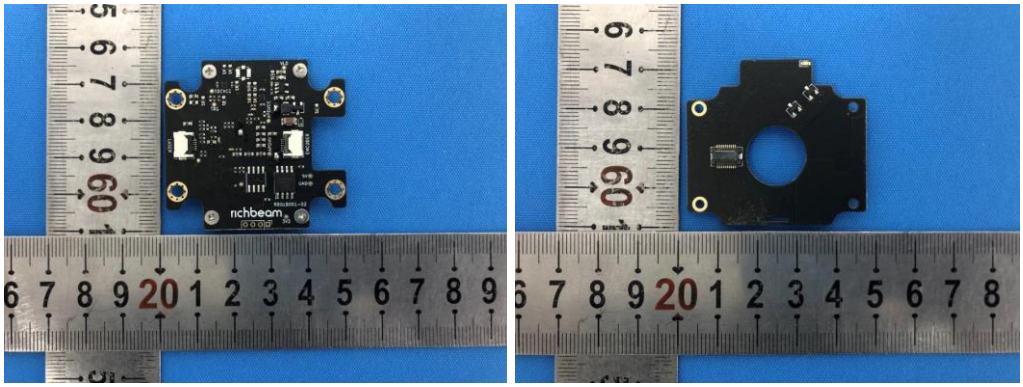
EN IEC 61496-1			
Clause	Requirement + Test	Result - Remark	Verdict
	The requirements set out in this document are such that a high dependence is placed on test equipment and expert analysis. In order to make a proper design review of electro-sensitive protective equipment to this document, the evaluation and testing in accordance with the requirements should be reviewed by an appropriate independent organization.	informative	-

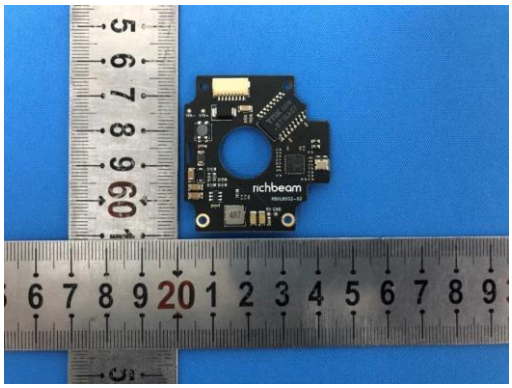
Photo Documents







--- End of Test Report ---